SUMMARY · MK 5 DECISION MAKING & NEGOTIATION · WEEK 7

Analytic Hierarchy Process (AHP)

Saaty's AHP: hierarchy of goal–criteria–alternatives, pairwise comparisons on a 1–9 scale, weights via normalisation/eigenvector, consistency index and ratio (CR ≤ 0.10), global synthesis, comparison with SMART, and criticisms (rank reversal).

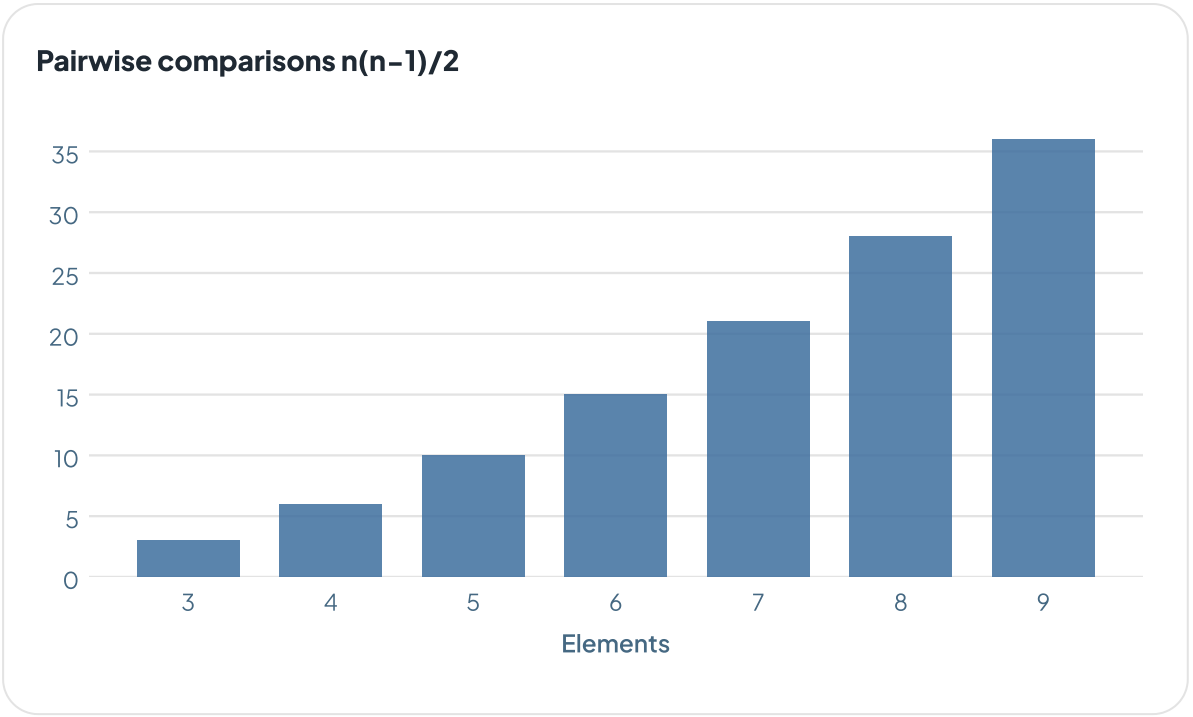
Material

1 Hierarchy and Saaty's scale

Steps: hierarchy, pairwise criteria matrix, weights and consistency, pairwise alternatives per criterion, global synthesis, sensitivity. Scale 1 (equal) to 9 (extreme); reciprocal matrix; n(n–1)/2 judgements.

2 The supplier hierarchy Deep dive





3 Computing weights and consistency

Example criteria quality, price, delivery with judgements 3, 5, 3: weights 0.633 / 0.260 / 0.106; λmax ≈ 3.039, CI 0.019, CR 0.033 (≤ 0.10 is acceptable; RI n=3 0.58, n=4 0.90, n=5 1.12). Synthesis: supplier X 0.469, Y 0.288, Z 0.242.

4 AHP calculator and global synthesis Deep dive

Global supplier priorities



Totals: X 0.469 · Y 0.288 · Z 0.242.

X wins on quality, the heaviest criterion.

5 AHP vs SMART, and criticisms

SMART: direct ratings and swing weights — fast and transparent. AHP: pairwise judgements with a consistency check, good for qualitative criteria and groups (geometric mean), but many comparisons, scale issues and rank reversal.

6 Rank reversal in numbers, and combining group judgements

The best-known criticism of AHP is **rank reversal** (Belton & Gear, 1983): adding a new alternative — even a copy of an existing one — can flip the order of the old alternatives. Example with three equally weighted criteria (1/3 each) and relative scores normalised per criterion:

Raw score	Criterion 1	Criterion 2	Criterion 3

A	1	9	8
B	9	1	9
C	1	1	1
D (copy of B)	9	1	9

Global priority	A	B	C	D
Only A, B, C	0.451	**0.470**	0.079	—
After adding D	**0.365**	0.289	0.057	0.289

Before D exists, B beats A. After adding D (identical to B), A beats B. The cause: normalising per criterion shares out the score among all alternatives, so D's presence shrinks B's share on the criteria where B is strong. Logically, adding an irrelevant option should not change the choice between A and B.

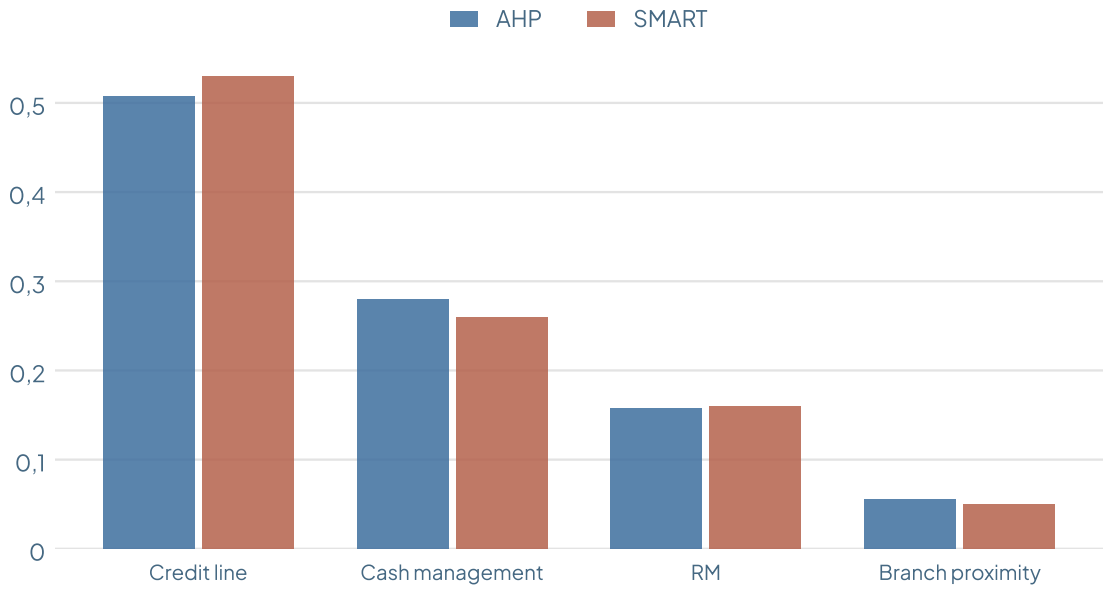
Ways to reduce the problem

(1) Use the *ideal* mode (divide by the highest score, not the sum) — available in AHP software. (2) Do not add clearly dominated alternatives. (3) For numerical criteria (price, time), use measured values directly instead of pairwise comparisons. (4) Always cross-check with another method (such as SMART) — as in the BUMDes exercise, where AHP and SMART weights were almost identical.

Combining group judgements. When several people give pairwise judgements, combine them with the **geometric mean**, not the arithmetic mean, so the reciprocal property is preserved. Example: three managers rate Quality vs Price as 3, 5 and 1/3. Geometric mean = $(3 \times 5 \times 1/3)^{1/3} = 5^{1/3} \approx \mathbf{1.71}$, and automatically Price vs Quality = 1/1.71 ≈ 0.58. The arithmetic mean (3 + 5 + 0.33)/3 = 2.78 would overstate it because it does not treat 3 and 1/3 symmetrically. If members' judgements differ widely, the combined number hides the conflict — discuss the reasons before averaging.

7 Answer: the BUMDes exercise with AHP Deep dive

BUMDes weights: AHP vs SMART




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λmax ≈ 4.008, CI 0.003, CR ≈ 0.003
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AHP weights nearly match SMART's, so any difference comes from how alternatives are scored.

Key takeaways

1. Goal → criteria → alternatives.
2. 1–9 scale, reciprocal.
3. Normalise columns, average rows.
4. CR ≤ 0.10.
5. Global = Σ weight × local.
6. Beware rank reversal.

Case studies

Case & worked answers

Exercise: the BUMDes case with AHP

Redo the BUMDes bank choice with AHP (example judgements given), check CR with RI 0.90, compare banks pairwise per criterion, and compare the result with SMART — methods shape outcomes, so sensitivity is essential.

Worked examples

Problem 1 · Medium · **Bobot AHP (matriks konsisten)**

Kriteria A, B, C. A vs B = 2, A vs C = 4, B vs C = 2. Hitung bobot dengan normalisasi kolom.

1. Matriks: baris A (1, 2, 4); B (1/2, 1, 2); C (1/4, 1/2, 1).
2. Jumlah kolom: 1,75; 3,5; 7.
3. Normalisasi baris A: 1/1,75 = 0,571; 2/3,5 = 0,571; 4/7 = 0,571 → rata-rata 0,571.
4. Dengan cara sama: B = 0,286; C = 0,143.
5. Semua kolom identik → sangat konsisten (CR = 0), karena 2 × 2 = 4 (A/B × B/C = A/C).

Answer: A 0,571; B 0,286; C 0,143; CR = 0.

Problem 2 Challenging **Mendeteksi penilaian tidak konsisten**

Penilai menulis: A vs B = 3, B vs C = 3, tetapi C vs A = 2 (C lebih penting dari A). Apakah konsisten? (RI n = 3 adalah 0,58.)

1. Logikanya: A > B dan B > C, maka seharusnya A > C (sekitar 9). Tetapi penilai bilang C > A — bertentangan.
2. Jika dihitung, bobot menjadi hampir rata (A 0,37; B 0,33; C 0,30) dan λmax ≈ 4,01.
3. CI = (4,01 – 3) ÷ 2 = 0,50; CR = 0,50 ÷ 0,58 = 0,87.

Answer: CR ≈ 0,87 jauh di atas 0,10 → tidak konsisten. Minta penilai meninjau ulang perbandingan A vs C.

Practice questions

Try answering first. The answer key is below each question.

1. If A vs B = 5, B vs A = …

A. 5
B. −5
C. 1/5
D. 0

Answer: C. Reciprocal matrix.

2. Acceptable CR is…

A. ≤ 0.01
B. ≤ 0.10
C. ≤ 0.50
D. ≥ 1

Answer: B. Saaty.

3. For 6 criteria, comparisons needed…

A. 6
B. 12
C. 15
D. 36

Answer: C. 6×5/2.

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4. Rank reversal means…

A. Inverted weights
B. Adding an irrelevant option can change the ranking
C. Negative CR
D. Non-reciprocal

Answer: B. A key criticism.

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5. RI for n = 4 is…

A. 0.58
B. 0.90
C. 1.12
D. 1.24

Answer: B. Used in the BUMDes exercise.

Sources

- [for Students] 07 Decision Analysis: AHP (MK5) — slide berupa gambar; contoh disusun ulang
- Saaty, The Analytic Hierarchy Process
- Goodwin & Wright, Bab 16 (AHP)